

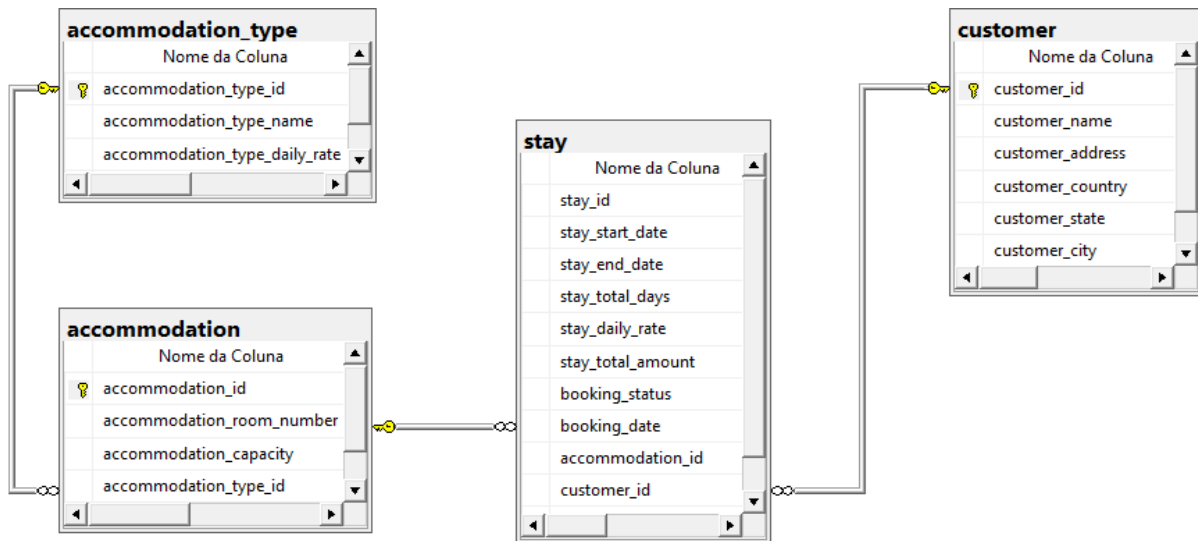
HotelAndStars

1. Objective.

Build an Entity-Relationship Model (ERM) and, from it, a Dimensional Model of the business model of the fictional company HotelAndStars.

2. Entity-Relationship Model (ERM) - SQL Server Management Studio (SSMS).

Entity-Relationship Model (ERM) created:



Generated files available at '1. Models/1. Relational/'.

3. Dimensional Model - SQL Server Management Studio (SSMS).

1. To create the analytical database structure model (OLAP) based on the transactional database (OLTP), using the Star Schema model (the most common model for Business Intelligence), the following transformations were made:

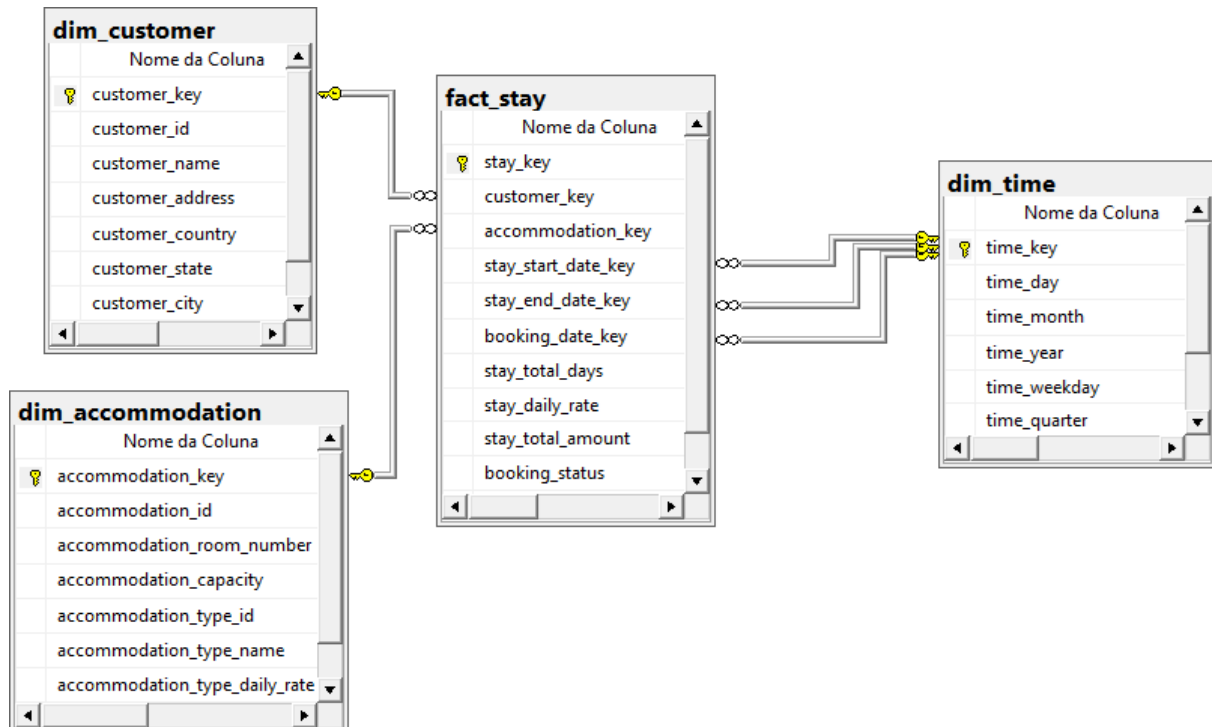
- **Customer ('who?'):** *dim_customer*, which contains the attributes of the *customer* table from the OLTP environment, plus the addition of *validity_start_date* and *validity_end_date* for versioning;
- **Accommodation ('where?'):** *dim_accommodation*, which contains the attributes of the *accommodation* e *accommodation_type* (used only for classification, therefore incorporated) from the OLTP environment, plus the addition of *validity_start_date* and *validity_end_date* for versioning.

2. The time dimension was created ('when?'): *dim_time*, which will have the attributes *time_day*, *time_month*, *time_year*, *time_weekday* and *time_quarter*, to enable dimension hierarchies in analyses;

3. In the OLAP environment, the fact table *fact_stay* consolidated the *stay* table. It holds its Surrogate Key (SK) as the Primary Key (PK), the dimension SKs as its Foreign Keys (FK), and the attributes of the *stay* table as metrics (the date attributes became FKs, as they are dates that mark when the fact occurred).

- The granularity of the *fact_stay* fact table was defined at the stay-per-customer level, ensuring the highest level of detail possible.

4. Dimensional Model created:



Generated files available at '1. Models/2. Dimensional/'.